DESCRIPTION OF COURSEWORK

Course Code	AIT201
Course Name	Applied Machine Learning
Lecturer	Dr. Mas Ira Syafila Binti Mohd Hilmi Tan
Academic Session	202604
Assessment Title	Final Assessment (Group Project)

A. Introduction/ Situation/ Background Information

This course introduces the principles of various machine learning (ML) techniques. It covers topics such as the formulation of learning problems and various applications of ML. The concepts are practiced using the tools from Python, Pandas, Seaborn, Scikit-Learn and TensorFlow, in exercises and a project.

In this project, students work in teams to apply machine learning knowledge and skills to solve a real-world problem based on one selected scenario. Students will gain practical experience in working with data, building and evaluating models, and interpreting results. The project aims to help students understand how to select and apply appropriate machine learning techniques, interpret results meaningfully, and present findings effectively.

Technical Skills Required:

- Python Programming: Basic to intermediate level of proficiency.
- Machine Learning: Familiarity with ML concepts and techniques.
- Data Preprocessing: Knowledge of data preprocessing
- Exploratory Data Analysis (EDA): Skills to visualize and interpret data patterns.
- Model Evaluation: Understanding of performance metrics to evaluate models.
- Deep Learning (optional): For more advanced techniques such as neural network-based models.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 3 Build the application of machine learning method to solve practical problem.

CLO 4 Perform the skills of teamwork during the development of the machine learning application.

[CLO2, C3, PLO3], [CLO4, C3, PLO8]

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
   iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.
2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

This is a group project for AIT201 (Applied Machine Learning) in which each team will apply machine learning techniques to solve a real-world data problem. Students are required to form a group consisting of **SIX (6)** to **SEVEN (7)** members each.

Each group must select **ONE scenario** from the provided options in **section F** and complete the entire machine learning workflow. This includes problem formulation, data preprocessing, exploratory data analysis, model development, evaluation, and interpretation of results. Groups are expected to work as a machine learning team, with clearly defined roles and contributions from each member.

The evaluation of this project is based on **TWO (2)** components: Component 1: Project Development and Component 2: Project Report, Presentation, Demonstration and Participation.

Submission Requirements:

- Submit a **single PDF report** and your **Jupyter Notebook (.ipynb) file**, compressed into a single **.zip file** on Moodle The report needs to be checked using Turnitin, and the one-page similarity report must be included. The similarity index should be below 20%; otherwise, it may result in mark deduction.
- The deadline for this assessment is **Thursday, 25th June 2026, by 11:00 PM**. Submissions after this date will receive a 0 Grade.
- Only the group leader is responsible for submitting the files on behalf of the group. The submission file must follow the naming format: *Group Number_Project Name.zip*.
- If your file is too large to be submitted in Moodle, submit in Moodle your Word file containing the OneDrive link.

The presentation will be conducted during **Week 13 to Week 14** during lecture or lab sessions if necessary. Each group will be given **15 minutes** for presentation, followed by a **5-minute Q&A session**. No extra time will be allocated, and all members must participate. A demonstration of the developed model is required. Peer assessment may be conducted during the submission week to evaluate individual contribution and ensure fairness among group members. The results may be used to adjust individual marks where necessary.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Project Development (code, implementation)	20
2.	Project Report, Presentation, Demonstration and Participation	30
	TOTAL	**50**

F. Tasks

Each group is required to select **ONE** scenario and complete the project based on the given context. A scenario may involve one or more machine learning problem types, and additional approaches may be applied where appropriate. The dataset used must be appropriate for the selected scenario, and its source must be clearly stated. Each group is expected to work as a machine learning team, with every member taking on at least one defined role. The roles and contributions of each member must be clearly documented in the report and presented during the presentation.

1. Component 1: Project Development (20%) [CLO3, CLO4]

Students are required to apply the knowledge and skills acquired in this course to develop a complete machine learning project based on the selected scenario. The project should demonstrate a clear and structured workflow, including data preparation, model development, and evaluation. Students must submit Python code files (e.g., Jupyter Notebook) that clearly show the complete implementation process from data preprocessing to model evaluation.

The deadline for this component is **Thursday, 25th June 2026, by 11:00 PM**. Only the team leader is required to submit the project on behalf of the group through the Turnitin system. The submission must include **a single PDF report** and the **Jupyter Notebook (.ipynb) file**, compressed into **a single .zip**. Please name the submission file as *Group Number_Project Name.zip* (e.g., Group2_RetailAnalysis.zip). Late or missing submissions will result in **zero** marks for this component.

Students are expected to submit a fully completed project that demonstrates the following:

a) Code quality, structure, and organization (10%)
b) Implementation of relevant machine learning algorithms and methods appropriate to the selected scenario (10%)

2. Component 2: Project Report, Presentation and Demonstration (30%) [CLO3, CLO4]

Each group is required to prepare a project report, presentation, and demonstration based on the developed project. The report must describe the project objectives, methodology, and results in a clear and well-organized manner. The report should not exceed **thirty (30) pages**, excluding the Appendix and Front Pages. All submissions are through Moodle platform. Late or missing submissions will result in zero marks for this component.

Your report must include the following sections:

a) **Title:** Provide a clear and descriptive title that represents the focus of your project.
b) **Abstract:** Provide a concise summary of your project, including the main objective, dataset used, methods applied, key results, and conclusions.
c) **Introduction:** Present an overview/background of the chosen scenario, its significance, and relevance to machine learning.
d) **Literature Review:** Present a discussion of relevant studies, research papers, or real-world applications related to your topic. This section should be approximately **one to two pages** in length and must include proper citations and references. Students are expected to review **at least two to three related studies or applications** that use machine learning techniques similar to their selected scenario. Use reliable sources such as Google Scholar, IEEE Xplore, ScienceDirect, or SpringerLink, and cite all references using **APA format**.
e) **Methodology:** Describe the steps and techniques used for data preprocessing, exploratory data analysis, model selection, and evaluation. Include a description of the dataset used.
f) **Results and Discussion:** Present and discuss your findings with supporting visualizations and performance metrics.
g) **Conclusion:** Summarize the key findings, achievements, limitations, and possible improvements for future work.
h) **References:** Provide a list of all sources cited in the report using **APA format**.
i) **Appendices:** The appendices may include supporting materials such as additional figures, tables, or supplementary code (optional). **Roles and Contributions** of each member must be included in the appendices (compulsory). This section should clearly describe the role assigned to each member, responsibilities, and contributions to the overall project.

Students are required to prepare a 15-minute presentation including a demonstration of the developed model, followed by a 5-minute Q&A session. Presentations will be conducted **during Week 13 to Week 14** during class hours or lab sessions if necessary. Each team must present within the allocated time and ensure that the key aspects of the project are clearly communicated.

The evaluation for Component 2 includes:

- Project Report (15%)
- Presentation, Demonstration and Participation (15%)

3. Project Scenarios (Choose ONE option)

Option A: Medical Data Analysis

Situation:

Assume you are part of a machine learning team in a healthcare organization. Your team is responsible for analyzing patient or clinical data to support decision-making, improve diagnosis, or enhance healthcare services.

Suggested Dataset: Medical or healthcare datasets (e.g., diabetes, heart disease, patient records) from any relevant dataset. The dataset may be obtained from public sources or other appropriate sources.

Tasks:

1. Identify and justify appropriate machine learning problem(s) based on the dataset.
2. Perform data preprocessing and exploratory data analysis (EDA).
3. Develop and compare at least two machine learning models.
4. Evaluate models using suitable evaluation metrics.
5. Interpret results and discuss their relevance to the medical context.

Optional: Additional analysis may be conducted to uncover meaningful patterns or groupings in the data.

Option B: Manufacturing & Quality Analysis

Situation:

Assume you are part of a machine learning team in a manufacturing company. Your team is tasked with analyzing production data to improve efficiency, detect defects, and enhance quality control processes.

Suggested Dataset: Manufacturing or industrial dataset (e.g., defect detection, sensor data, production logs) from any relevant dataset. The dataset may be obtained from public sources or other appropriate sources.

Tasks:

1. Identify and justify appropriate machine learning problem(s) based on the dataset.
2. Perform data preprocessing and exploratory data analysis (EDA).
3. Develop and compare at least two machine learning models.

4. Evaluate models using suitable evaluation metrics.
5. Interpret findings and discuss how they can improve manufacturing processes.

Optional: Additional analysis may be conducted to uncover meaningful patterns or groupings in the data.

Option C: Retail/ Sales Analytics

Situation:

Assume you are part of a data team in a retail or sales-based company. Your team is analysing customer and transaction data to support business decisions, improve sales performance, and understand customer behaviour.

Suggested Dataset: Any relevant dataset related to retail, sales, or customer transactions (e.g., product data, customer behaviour, or sales records). The dataset may be obtained from public sources or other appropriate sources.

Tasks:

1. Identify and justify appropriate machine learning problem(s) based on the dataset.
2. Perform data preprocessing and exploratory data analysis (EDA).
3. Develop and compare at least two machine learning models.
4. Evaluate models using suitable evaluation metrics.
5. Interpret findings and discuss their business implications.

Optional: Additional analysis may be conducted to uncover meaningful patterns or groupings in the data.

APPENDIX 1

MARKING RUBRICS

Component 1: Project Development

		Score and Descriptors				
No.		**Poor**	**Average**	**Excellent**	**Weight (%)**	**Mark**
		0 - 4	**5 - 7**	**8 - 10**	**10**	
1	Code quality, structure, and organization	Codes are poorly structured, not fully functional, and not documented.	Codes are mostly functional with acceptable structure and some documentation.	Codes are fully functional, well-structured, and clearly documented.		
		0 - 4	**5 - 7**	**8 - 10**	**10**	
2	Implementation of machine learning methods appropriate to the selected scenario	Incomplete or inappropriate implementation of methods.	Appropriate implementation of methods with reasonable results.	Complete and well-justified implementation with enhancements		
				Subtotal	**20**	

Component 2: Project Report + Presentation, Demonstration & Participation

		Score and Descriptors				
No.		**Poor**	**Average**	**Excellent**	**Weight (%)**	**Mark**
		0 - 5	**6 - 10**	**11 – 15**	**15**	
1	Project Report	Poor writing quality, poor formatting, and lack	Satisfactory writing quality with adequate structure	Well-written, clearly structured, and demonstrates		

		of technical content.	and technical content.	strong technical understanding and analysis.		
		0 - 1	**2 - 3**	**4 - 5**	**5**	
2	Slides, technical content, demonstration and response to questions (Group)	Poor slides, unclear structure, weak technical content. Demonstration is incomplete or not functioning. Unable to respond to questions.	Acceptable slides with basic structure and content. Demonstration is mostly functional. Able to answer basic questions with some limitations.	Clear, well-structured slides with strong technical content. Demonstration is smooth and fully functioning. Able to respond to questions confidently and accurately.		
		0 - 1	**2 - 3**	**4 - 5**	**5**	
3	Presentation delivery (Individual)	Unclear delivery, poor communication, lacks confidence.	Understandable delivery with some clarity.	Clear, confident, and well-communicated delivery.		
		0 - 1	**2 - 3**	**4 - 5**	**5**	
4	Participation (Individual)	No or minimal submission.	At least 50% of lab submissions completed.	At least 80% of lab submissions completed.		
				Subtotal	**30**	
				Grand Total	**50**	

Note to students: Please include the marking rubric when submitting your coursework